

CHANDIGARH UNIVERSITY APEX INSTITUTE OF TECHNOLOGY



EXPERIMENT 12

**Subject: Advanced Database Management
System**

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Question 1:

```
-----MEDIUM LEVEL PROBLEM-----  
--REQUIREMENTS: DESIGN A TRIGGER WHICH:  
--1. WHENEVER THERE IS A INSERTION ON STUDENT TABLE THEN, THE CURRENTLY INSERTED OR DELETED  
--ROW SHOULD BE PRINTED AS IT AS ON THE OUTPUT CONSOLE WINDOW.
```

Solution:

```
--SOLUTION  
CREATE OR REPLACE FUNCTION fn_student_audit()  
RETURNS TRIGGER  
LANGUAGE plpgsql  
AS  
$$  
BEGIN  
    IF TG_OP = 'INSERT' THEN  
        RAISE NOTICE 'Inserted Row -> ID: %, Name: %, Age: %, Class: %',  
            NEW.id, NEW.name, NEW.age, NEW.class;  
        RETURN NEW;  
  
    ELSIF  
        TG_OP = 'DELETE' THEN  
        RAISE NOTICE 'Deleted Row -> ID: %, Name: %, Age: %, Class: %',  
            OLD.id, OLD.name, OLD.age, OLD.class;  
        RETURN OLD;  
    END IF;  
  
    RETURN NULL;  
END;  
$$
```

Question 2:

```
/*
Requirements: DESIGN A POSTGRESQL TRIGGERS THAT:

Whenever a new employee is inserted in tbl_employee, a record should be added to tbl_employee_audit like:
"Employee name <emp_name> has been added at <current_time>"

Whenever an employee is deleted from tbl_employee, a record should be added to tbl_employee_audit like:
"Employee name <emp_name> has been deleted at <current_time>"

The solution must use PostgreSQL triggers.
*/
```

Solution:

```
CREATE TABLE tbl_employee (
    emp_id SERIAL PRIMARY KEY,
    emp_name VARCHAR(100) NOT NULL,
    emp_salary NUMERIC
);

CREATE TABLE tbl_employee_audit (
    sno SERIAL PRIMARY KEY,
    message TEXT
);

CREATE OR REPLACE FUNCTION audit_employee_changes()
RETURNS TRIGGER
LANGUAGE plpgsql
AS
$$
BEGIN
    IF TG_OP = 'INSERT' THEN
        INSERT INTO tbl_employee_audit(message)
        VALUES ('Employee name ' || NEW.emp_name || ' has been added at ' || NOW());
        RETURN NEW;

    ELSIF TG_OP = 'DELETE' THEN
        INSERT INTO tbl_employee_audit(message)
        VALUES ('Employee name ' || OLD.emp_name || ' has been deleted at ' || NOW());
        RETURN OLD;

    END IF;

    RETURN NULL;
END;
```

```
CREATE TRIGGER trg_employee_audit
AFTER INSERT OR DELETE
ON
tbl_employee
FOR EACH ROW
EXECUTE FUNCTION audit_employee_changes();
```

```
456
457 INSERT INTO tbl_employee(emp_name, emp_salary) VALUES ('Aman', 50000);
458
459 -- Delete an employee
460 DELETE FROM tbl_employee WHERE emp_name = 'Aman';
461
462 -- Check audit log
463 SELECT * FROM tbl_employee_audit;
```

Data Output

Messages

Notifications

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